

Computer Vision SS26 - Exercise 6

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This exercise will be discussed the week of 15th of June.

1 Introduction

The following exercises cover Structure from Motion (SfM), Bundle Adjustment, and Stereo Disparity.

2 Exercises

Exercise 1: Structure from Motion in 1D

Imagine a simplified 2D world where cameras only have a 1D image sensor (a line). The cameras are located along the X-axis and look down the Z-axis. The projection of a 2D point $\mathbf{X} = (X, Z)$ onto a camera located at position t on the X-axis is given by a 1D pinhole projection formula:

$$u = \frac{X - t}{Z}$$

We have two cameras. Camera 1 is fixed at the origin ($t_1 = 0$). Camera 2 is at an unknown position $t_2 = t$. We observe two unknown points, $\mathbf{P}_A = (X_A, Z_A)$ and $\mathbf{P}_B = (X_B, Z_B)$. The observed image coordinates are:

- Point A: $u_{1A} = 2$ in Camera 1, and $u_{2A} = 1$ in Camera 2.
 - Point B: $u_{1B} = -1$ in Camera 1, and $u_{2B} = -2$ in Camera 2.
1. Formulate the system of projection equations for these observations. How many equations and how many unknowns are there?
 2. Explain why this system cannot be solved to a unique solution. What fundamental property of Structure from Motion does this represent?
 3. Fix the scale ambiguity by assuming the distance between the cameras is exactly $t = 1$. Solve the system for the 2D coordinates of both points.

4. Calculate the exact sum of squared reprojection errors if our initial guess was $t = 1$, $\mathbf{P}_A = (2.5, 1.5)$, and $\mathbf{P}_B = (-1.5, 1.2)$. The reprojection error is the sum of squared differences between the observed and predicted image coordinates: $\sum (u_{obs} - u_{pred})^2$.

Exercise 2: Bundle Adjustment

In large-scale Structure from Motion, bundle adjustment is used to jointly optimize camera parameters and 3D point locations by minimizing the reprojection error.

1. Given n calibrated cameras and m 3D points. Each camera is described by its extrinsics (a rotation and a translation). How many total parameters need to be optimized?
2. If every point is visible in every camera, how many individual error terms (residuals) are accumulated in the sum of squared reprojection errors? Remember that a 2D image point provides two coordinate values (u, v) .
3. Write down the objective function $g(\mathbf{X}, \mathbf{R}, \mathbf{T})$ for Bundle Adjustment. Briefly explain the role of the indicator variable $w_{i,j}$.
4. Why is this optimization called a *non-linear* least squares problem?

Exercise 3: Stereo Disparity and Depth

For dense stereo matching, we usually assume a rectified stereo camera setup.

1. Why is image rectification performed before stereo matching? How does it simplify the search for corresponding points?
2. Given a stereo setup with a baseline $b = 0.2$ meters and a focal length $f = 800$ pixels. State the formula relating depth z to disparity d .
3. Calculate the depth of a point if its disparity is $d = 40$ pixels.
4. Block matching uses metrics like Normalized Cross-Correlation (NCC) to find correspondences. What is a common problem with block matching in textureless regions and at object boundaries (half-occlusions)?